

Machine Learning for Cross-Sectional Stock Return Prediction

Preliminary Project Report

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July 21, 2026

Abstract

This paper compares linear, regularized, dimension-reduction and tree-based methods for predicting next-month stock excess returns. The models are evaluated using a chronological design that separates model selection from a 2020–2025 test period. Partial Least Squares produces the lowest preliminary test-period mean squared error and an out-of-sample R^2 of approximately 0.18%. Random Forest and a parsimonious three-variable OLS model also slightly outperform the historical-mean benchmark. Elastic Net selects no nonzero coefficients and effectively reproduces the benchmark, while the single Decision Tree performs substantially worse. The results illustrate that machine-learning flexibility does not automatically translate into reliable stock-return forecasts and that small improvements must be interpreted cautiously. These estimates are preliminary: a project audit identified target continuity, macro-timing and workflow-documentation issues that will be corrected before a complete clean rebuild.

Keywords: stock returns, machine learning, Partial Least Squares, Random Forest, out-of-sample prediction

1 Introduction

Expected stock returns are central to asset pricing, portfolio allocation and risk management. Predicting them is nevertheless difficult because realized monthly returns contain substantial idiosyncratic variation and because predictive relationships may change across market regimes. A model that fits historical observations well may therefore perform poorly when applied to future data.

Machine-learning methods are potentially attractive in this setting because they can process many correlated predictors, impose regularization and capture nonlinearities or interactions. At the same time, their flexibility increases the risk of overfitting. A meaningful evaluation must consequently be based on genuinely out-of-sample observations and must compare complex models with a simple benchmark.

This paper asks:

Can linear dimension-reduction, regularized regression and tree-based methods improve next-month stock excess-return forecasts relative to a historical-mean benchmark?

The study compares six approaches under a common information set and evaluation framework: the historical mean, OLS with three economically motivated predictors, Partial Least Squares,

Elastic Net, a Decision Tree and a Random Forest. Model selection is performed chronologically, while 2020–2025 is used as the preliminary test period in the current implementation.

The study is related to the predictive asset-pricing literature, particularly the use of high-dimensional firm characteristics in Gu et al. (2020). It also follows the benchmark-oriented perspective of Welch and Goyal (2008), who show that apparently sophisticated return-prediction models often struggle to outperform simple historical forecasts.

The contribution is not the introduction of a new algorithm. Instead, the paper provides a transparent and reproducible comparison of models with different levels of flexibility. It places particular emphasis on negative findings, model instability and the distinction between statistical ranking and economically meaningful predictability.

This document is a preliminary project report. It describes the current implementation and reports its existing outputs transparently, but it does not present them as final empirical evidence. The project audit identified a small number of consequential timing and workflow issues. These are documented below and will be corrected before the output directory is deleted and the entire project is rebuilt from the permanent inputs.

2 Data

2.1 Sources and observation unit

The observation unit is a stock-month. The stock universe is based on a locked current-and-historical S&P 500-style ticker list. Daily price and volume information comes from Yahoo Finance. Market index and VIX series are used to construct aggregate market and volatility variables. Monthly Fama–French data provide the risk-free rate used to calculate excess returns. Annual Compustat data supply accounting characteristics and industry classifications. Welch–Goyal variables provide aggregate predictive states.

Daily price histories begin before the modeling period so that long-window characteristics can be constructed. The current monthly modeling sample begins in January 1990. The held-out test period covers January 2020 through December 2025.

For stock i at month t , the intended target is the excess return in the immediately following calendar month:

$$y_{i,t+1} = r_{i,t+1} - RF_{t+1}, \quad (1)$$

where $r_{i,t+1}$ is the stock return and RF_{t+1} is the one-month risk-free rate.

2.2 Timing and preprocessing

Predictors are constructed using information available by the end of month t . Momentum measures exclude the return being predicted. Annual Compustat observations are made available six months after fiscal year-end and are merged backward to the most recent report available for each ticker-month. This conservative lag reduces, but cannot fully eliminate, the risk of look-ahead bias because exact filing dates are unavailable.

The project audit confirms that stock-level feature construction and the six-month Compustat availability rule are largely chronological. It also finds two timing issues that must be corrected

before final estimation. First, the current ticker-level lead operation generally constructs the intended target, but in 26 retained observations the next available ticker observation is not the immediately following calendar month. These nonconsecutive transitions will be removed. Second, the Welch–Goyal macro variables are currently merged in the same month. An explicit one-month availability lag will be imposed so that the final rebuild uses only macro information available at the forecast date.

Missing predictor observations are replaced using the contemporaneous cross-sectional median. If an entire predictor is missing within one month, the remaining values are filled with zero. Predictors are then converted to within-month cross-sectional ranks on the interval $[-1, 1]$:

$$x_{i,t}^{\text{rank}} = 2 \frac{\text{rank}(x_{i,t}) - 1}{N_t - 1} - 1. \quad (2)$$

This transformation reduces sensitivity to units and extreme predictor values while preserving the relative ordering of firms within each month.

2.3 Predictor set

The current design contains 499 predictors. The base variables represent major economic groups:

- size, valuation, profitability, investment and leverage;
- momentum, reversals and price trends;
- volatility, downside risk, beta and idiosyncratic volatility;
- liquidity and trading activity;
- industry indicators;
- market, VIX and Welch–Goyal aggregate states;
- interactions between firm characteristics and aggregate states.

The resulting panel is high-dimensional and highly correlated. This provides the motivation for regularization, dimension reduction and nonlinear methods.

2.4 Chronological sample design

The data are divided chronologically:

- training period: 1990–2014;
- validation period: 2015–2019;
- preliminary test period: 2020–2025.

Hyperparameters are selected using annual expanding-window folds within the training period. Fixed and optimized specifications are compared using the validation period. Once the specifications are selected, the training and validation observations are combined into a development sample, each model is refitted once and evaluated on the preliminary test period. The project audit also identified inconsistencies between the written description of the tuning periods and selection rule and their implementation. The exact chronology and selection rule will therefore be reconciled and documented before the clean rebuild.

Random cross-validation is not used because it would allow future market observations to influence earlier forecasts. The chronological design more closely represents the information available to an investor in real time.

3 Methodology

3.1 Forecasting models

The general forecasting problem is

$$y_{i,t+1} = f(\mathbf{x}_{i,t}) + \varepsilon_{i,t+1}, \quad (3)$$

where the function $f(\cdot)$ differs across models.

Historical Mean. The benchmark predicts the mean target observed in the permitted estimation sample. It ignores firm characteristics and represents the performance of a simple constant forecast.

OLS-3. The parsimonious OLS specification uses size, book-to-market and 12-month momentum. These variables offer an interpretable reference between the constant benchmark and the high-dimensional models.

Partial Least Squares. PLS constructs supervised latent components from the 499 predictors. Unlike unsupervised dimension reduction, the components are chosen to be related to the target. PLS is particularly relevant when predictors are numerous and strongly correlated (Wold et al. 1984).

Elastic Net. Elastic Net combines L_1 and L_2 regularization:

$$\hat{\beta} = \arg \min_{\beta} \left[\frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \left(\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right) \right]. \quad (4)$$

The L_1 component permits variable selection, while the L_2 component stabilizes estimates when predictors are correlated (Tibshirani 1996; Zou and Hastie 2005).

Decision Tree. A regression tree recursively divides the predictor space and assigns a constant prediction to each terminal leaf. It can represent threshold effects and interactions but is sensitive to changes in the estimation sample.

Random Forest. Random Forest averages predictions across bootstrap trees and considers a random subset of predictors at each split (Breiman 2001). Averaging is intended to reduce the variance of an individual tree while preserving nonlinear predictive relationships.

3.2 Evaluation metrics

The primary criterion is monthly mean squared error:

$$\text{MSE}_{\text{month}} = \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{N_t} \sum_{i=1}^{N_t} (y_{i,t+1} - \hat{y}_{i,t+1})^2 \right]. \quad (5)$$

Errors are first averaged across firms and then across months, preventing months with more observations from receiving greater weight.

The report also presents RMSE, MAE, prediction-target correlation and out-of-sample R^2 :

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{i,t} (y_{i,t+1} - \hat{y}_{i,t+1})^2}{\sum_{i,t} (y_{i,t+1} - \hat{y}_{i,t+1}^{\text{HM}})^2}. \quad (6)$$

A positive value means that the model reduces squared forecast error relative to the historical mean. A negative value means that the model performs worse than the benchmark.

4 Preliminary Results

4.1 Overall predictive performance

Table 1: Preliminary test-period performance, 2020–2025

Rank	Model	MSE	RMSE	MAE	R_{OOS}^2	Correlation
1	PLS	0.0138	0.1172	0.0802	0.0018	0.0215
2	Random Forest	0.0138	0.1173	0.0801	0.0006	0.0256
3	OLS-3	0.0138	0.1173	0.0801	0.0005	–
4	Elastic Net	0.0138	0.1173	0.0801	1.3487×10^{-10}	–
5	Historical Mean	0.0138	0.1173	0.0801	-6.6012×10^{-11}	–0.1453
6	Decision Tree	0.0157	0.1248	0.0848	–0.1328	–

Notes: Models are ranked by monthly MSE. Correlation is undefined for constant or near-constant predictions. Positive R_{OOS}^2 indicates an improvement over the historical-mean benchmark.

PLS provides the lowest monthly MSE and the highest out-of-sample R^2 . Its R_{OOS}^2 of 0.00179 means that it reduces squared forecast error by approximately 0.18% relative to the historical mean. The result is positive, but economically small.

Random Forest ranks second with an out-of-sample R^2 of approximately 0.058%. OLS-3 ranks third with approximately 0.049%. The difference between these two models is minimal, indicating that the large increase in model complexity delivered by Random Forest produces only a small additional reduction in squared error relative to the parsimonious OLS specification.

Elastic Net and the historical mean produce almost identical results. The optimized Elastic Net contains no coefficients above the specified tolerance and therefore behaves as a constant predictor. This is a meaningful negative finding: after chronological validation, the regularized model found no sufficiently stable incremental signal among the 499 predictors.

The Decision Tree performs considerably worse than all other models. Its out-of-sample R^2 is approximately –13.3%, and its prediction-target correlation is negative. The individual tree appears to fit unstable splits that do not generalize to the test sample.

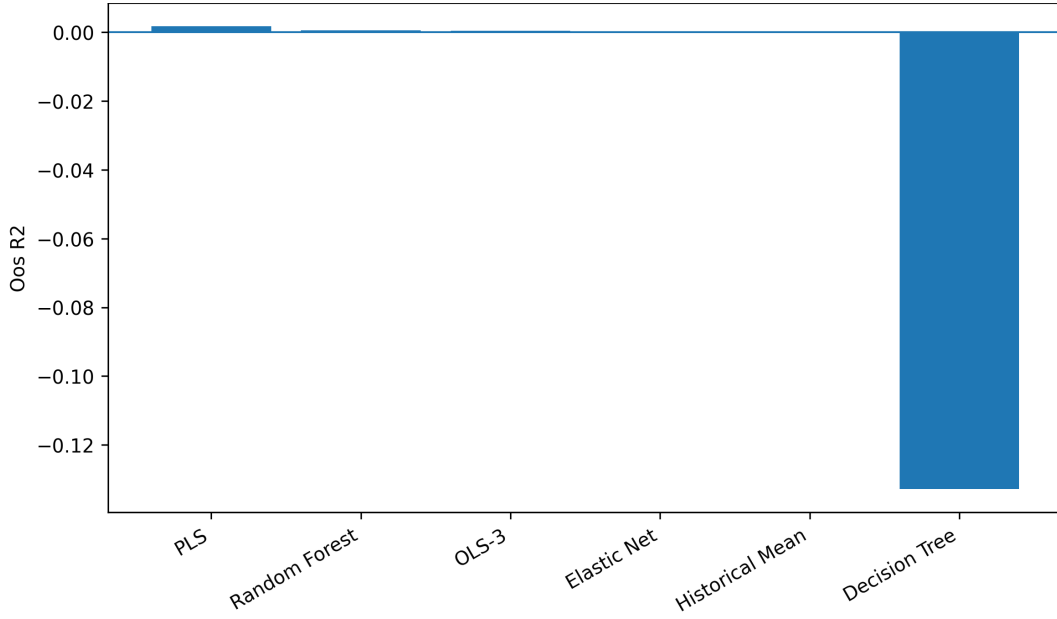


Figure 1: Preliminary test-period out-of-sample R^2 by model

Figure 1 emphasizes both the ranking and the limited absolute predictive power. PLS, Random Forest and OLS-3 outperform the benchmark, but all positive values remain close to zero. The Decision Tree is qualitatively different because its additional flexibility substantially increases forecast error.

The MAE ranking differs slightly from the MSE ranking. The historical mean and Elastic Net have the lowest MAE, while PLS has the lowest MSE. This suggests that PLS reduces some relatively large forecast errors, which receive more weight under squared loss, without reducing the typical absolute error.

4.2 Financial interpretation

The small positive out-of-sample R^2 values represent small reductions in forecast error relative to the historical-mean benchmark. They provide evidence of weak statistical predictability in the current project version, but they do not establish profitable trading. In particular, MSE does not account for portfolio turnover, transaction costs, liquidity, short-selling constraints or risk adjustment. Nor does it show that the cross-sectional ordering of forecasts can be converted into stable implementable positions. The present evidence should therefore be interpreted as a preliminary comparison of forecasting methods, not as a ready-to-use investment strategy.

4.3 Performance across test years

The preliminary pooled ranking does not imply stable performance in every year. PLS and Random Forest achieve positive yearly out-of-sample R^2 in half of the six test years. OLS-3 does so in one third of the years. The Decision Tree beats the benchmark in only one year.

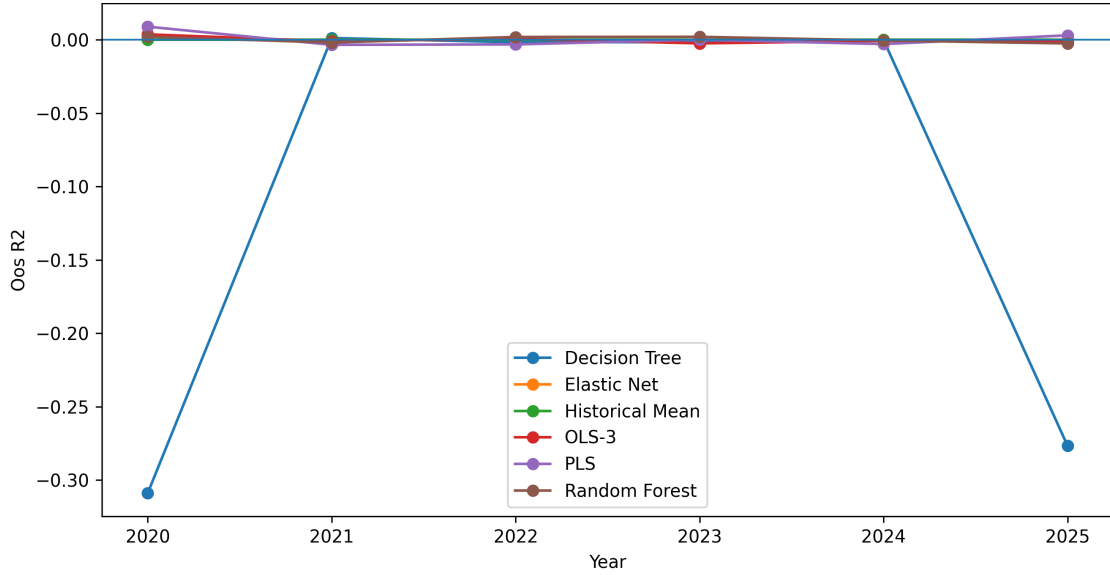


Figure 2: Out-of-sample R^2 by model and test year

The yearly results demonstrate that even the best overall model does not consistently outperform the historical mean. This instability is important because the test period contains only six years and includes unusually different market environments. The overall advantage of PLS should therefore be interpreted as an average test-period result rather than evidence of stable performance in every regime.

4.4 Forecast dispersion

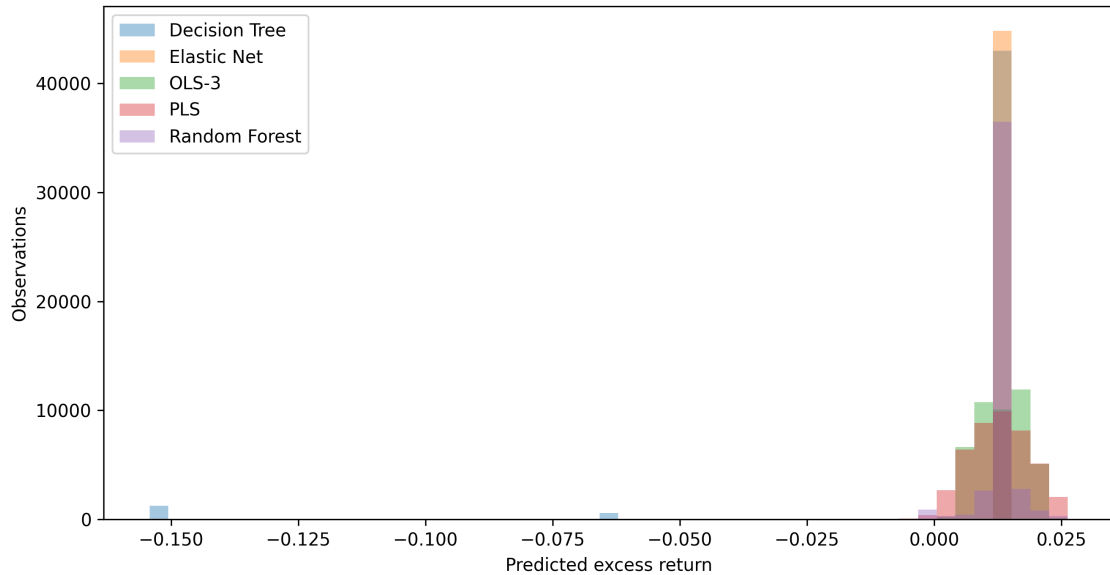


Figure 3: Distribution of preliminary test-period predictions

The prediction distributions help explain the model results. Elastic Net is absent as a meaningful distribution because it produces constant or near-constant forecasts. PLS and OLS generate relatively concentrated predictions, while the tree-based models produce more dispersed forecasts. Dispersion alone does not indicate predictive quality. The Decision Tree produces variable forecasts but performs poorly because those forecasts are systematically misaligned with realized returns.

4.5 Feature importance

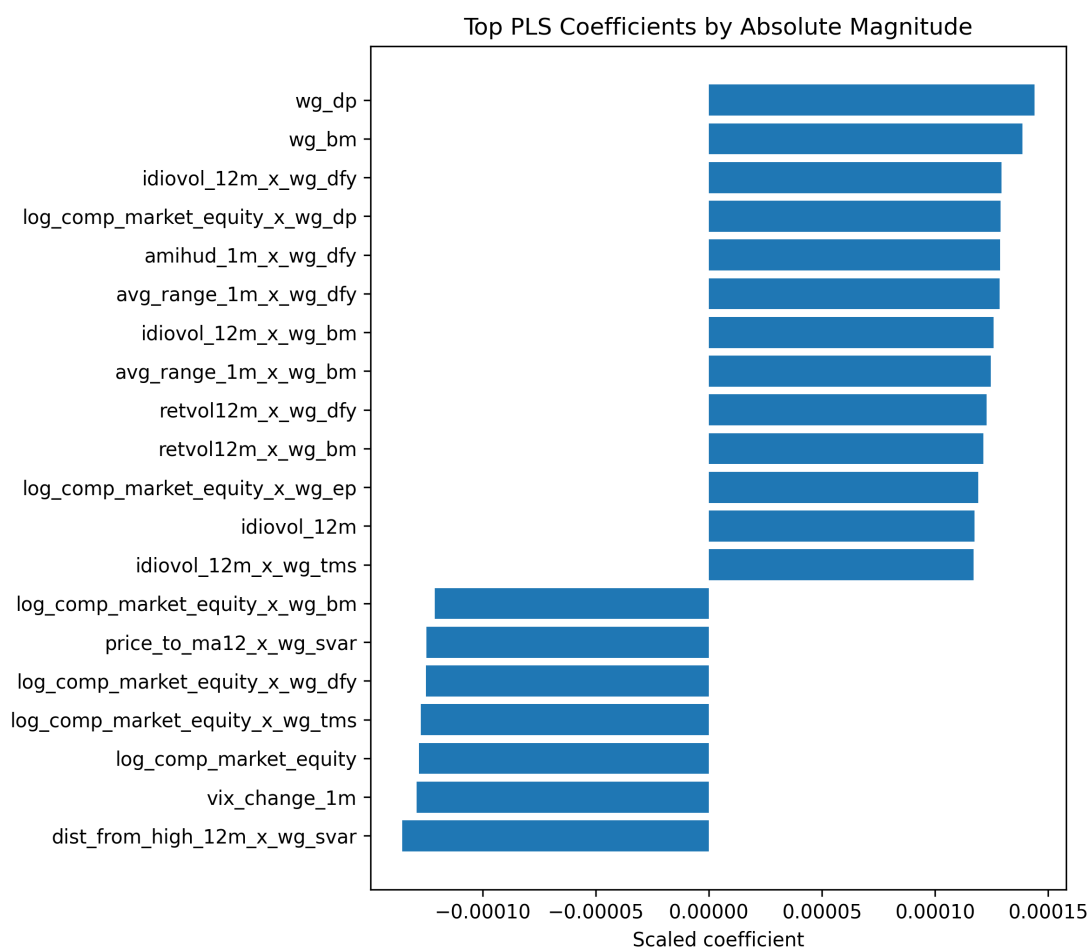


Figure 4: Largest PLS coefficients by absolute magnitude

PLS coefficients measure conditional predictive associations after preprocessing and scaling. Their magnitude should not be interpreted as a causal effect or as an estimated risk premium. Because PLS combines information through latent components, the coefficients reflect the joint high-dimensional prediction problem rather than isolated univariate relationships.

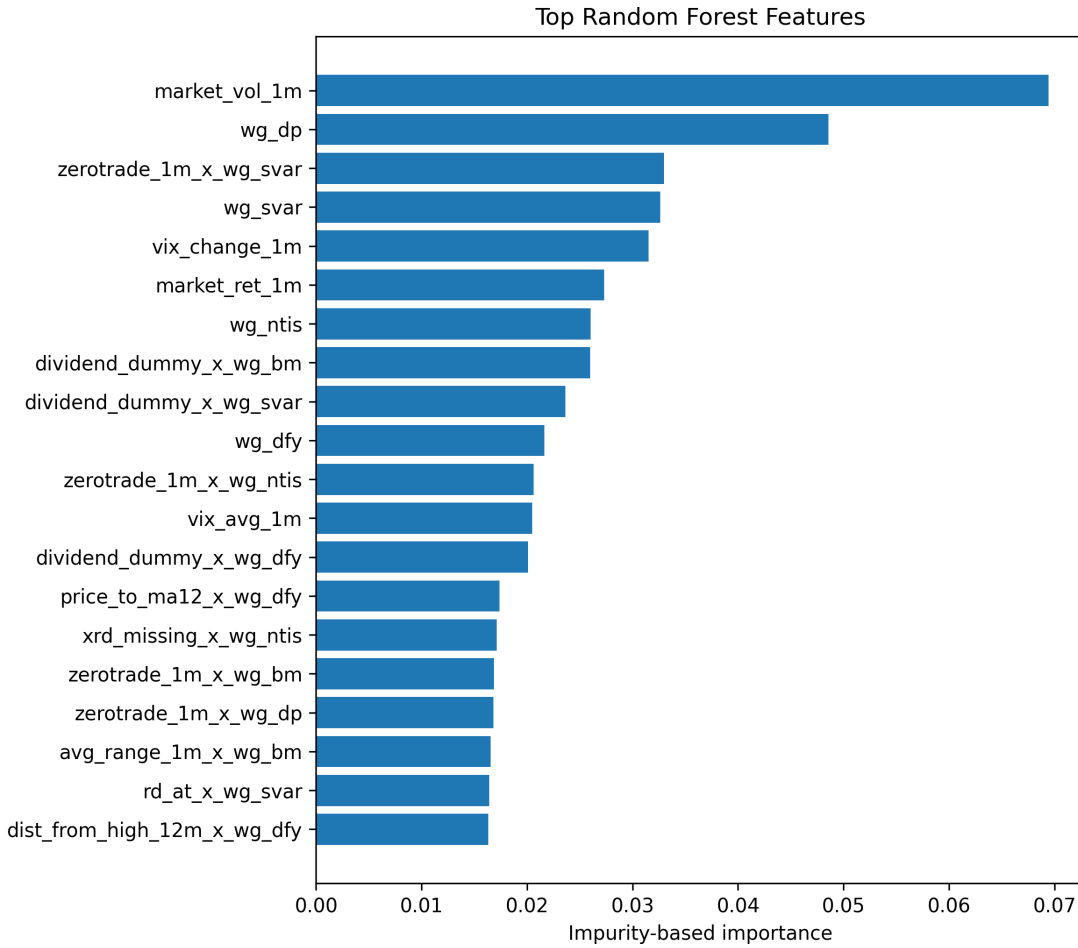


Figure 5: Top Random Forest predictors based on impurity importance

Random Forest importance records how strongly variables reduce squared error inside the fitted trees. It is useful for describing where the model obtains its predictive variation, but it has important limitations. Correlated variables can divide importance among themselves, and impurity importance can favor predictors with more available split points. The results should therefore be interpreted at the level of broad predictor groups rather than as a precise ranking of causal variables.

5 Critical Reflection

5.1 Predictive gains are small

The main limitation is the weak absolute predictive performance. Although PLS ranks first, its out-of-sample R^2 is only approximately 0.18%. Random Forest and OLS-3 improve on the benchmark by even less. Most variation in monthly stock returns therefore remains unexplained.

A ranking can still be statistically informative, but it should not be confused with strong forecasting ability. The results do not establish that the small differences between the leading models would remain stable in another period.

5.2 Flexible models do not automatically perform better

The results provide two important failures. First, the individual Decision Tree performs much worse than the historical mean. Its threshold splits are apparently too unstable for noisy monthly return data. Random Forest improves substantially by averaging across trees, illustrating the benefit of variance reduction.

Second, Elastic Net selects no nonzero coefficients. This does not imply a software error. Given the selected tuning parameters and chronological validation procedure, the model preferred complete shrinkage over unstable predictor effects. The result suggests that individual linear signals are weak relative to the noise in the target.

5.3 Implementation issues identified during the project audit

The project audit identified six implementation and reproducibility problems that prevent the current outputs from being described as final results:

- The ticker-level target lead contains 26 nonconsecutive transitions in which the next available observation is not the immediately following calendar month.
- Welch–Goyal macro variables are merged in the same month rather than with the intended explicit one-month availability lag.
- The documentation and implementation are inconsistent about the tuning periods and the model-selection rule.
- The top-level `run_pipeline.sh` became stale after the model code was reorganized and references stages that no longer match the active workflow.
- The output structure contains excessive duplicate and legacy artifacts, which makes it harder to identify the authoritative results.
- Software dependencies are not pinned to exact versions, limiting exact reproducibility across environments.

These issues do not make the current tables useless: they remain informative about model behavior and expose several robust negative findings. They do mean, however, that the numerical rankings must be treated as preliminary. Before the complete rebuild, the target continuity rule and macro lag will be corrected, the tuning design and documentation will be reconciled, the pipeline entry point will be repaired, dependencies will be pinned, and authoritative outputs will be separated from legacy artifacts. The existing output folder will then be deleted and regenerated from the permanent inputs in one clean chronological run.

5.4 Data limitations

The stock universe is not a fully point-in-time CRSP constituent universe. Ticker-based matching may be affected by symbol changes, mergers and incomplete historical membership records. Survivorship and universe-selection bias cannot be ruled out completely.

Yahoo Finance does not provide the security master, delisting returns and microstructure information available in CRSP. Adjusted price histories may also be revised. Compustat fundamentals are lagged by six months, but exact public filing dates are not used. The resulting timing design is conservative, though not perfectly point in time.

The use of annual Compustat information also means that accounting variables change less frequently than market variables. Some predictors are proxies for variables used in the published asset-pricing literature and may not reproduce their original definitions exactly.

5.5 Validation and specification risk

Annual expanding-window validation respects chronology and reduces direct leakage. Nevertheless, the project contains many researcher choices: predictor construction, interaction design, preprocessing, tuning grids and sample dates. These decisions may adapt indirectly to known characteristics of the data.

An untouched holdout period remains credible only if no model changes are made after inspecting its performance. The current 2020–2025 outputs have already been examined and are therefore reported only as preliminary evidence. The corrected workflow must be specified before the clean rebuild, and any subsequent specification search would require a genuinely new holdout period.

5.6 No economic trading evaluation

The project evaluates prediction error but does not construct portfolios. Consequently, it does not measure turnover, transaction costs, market impact, short-sale constraints or risk-adjusted investment returns. A small reduction in MSE cannot automatically be interpreted as an economically profitable strategy.

A useful extension would specify a portfolio rule before opening a new test sample. The evaluation could then report gross and net returns, volatility, Sharpe ratios, turnover and maximum drawdown. That analysis would have to be kept separate from the current test period to avoid using the same observations for model assessment and strategy design.

5.7 What I would do differently

With improved data access and additional time, the study could be strengthened in four ways:

- use a point-in-time CRSP/Compustat universe with permanent identifiers and delisting returns;
- use exact filing dates for accounting information;
- retain a longer untouched test period or evaluate several historical forecast origins;
- conduct a pre-specified portfolio evaluation with realistic trading costs.

Interpretability could also be improved by examining feature groups, repeated time-block importance and stability across different expanding windows rather than relying on a single definitive importance ranking.

6 Conclusion

This preliminary report examines whether machine-learning methods improve next-month stock excess-return prediction relative to a historical-mean benchmark. The models are selected chronologically and evaluated on a held-out 2020–2025 test period in the current implementation.

PLS performs best according to monthly mean squared error and produces an out-of-sample R^2 of approximately 0.18%. Random Forest and OLS-3 also slightly outperform the benchmark. Elastic Net selects no predictors and effectively collapses to the historical mean. The Decision

Tree performs substantially worse, showing that flexibility without sufficient variance control can be harmful.

The research question can therefore be answered cautiously. Some statistical and machine-learning methods improve on the historical mean, but the gains are small and unstable across years. The results do not demonstrate strong return predictability or a profitable investment strategy. Their principal lesson is that, in noisy financial prediction problems, disciplined chronology, simple benchmarks and honest reporting of failures are at least as important as model complexity. Because the audit identified target-continuity, macro-timing and workflow inconsistencies, these results are not the final project evidence. Those issues will be corrected before the output folder is deleted and the project is rebuilt cleanly from its permanent inputs.

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A Code and Reproducibility Overview

The repository implements the empirical workflow as a sequence of auditable stages rather than embedding source code in this report:

1. acquisition of permanent external inputs and cleaning of daily stock histories;
2. construction of monthly stock features and the next-month target;
3. merging of lagged fundamentals and aggregate macroeconomic variables;
4. final missing-value treatment and within-month rank normalization;
5. chronological hyperparameter tuning with expanding estimation windows;
6. preliminary test-period refitting and evaluation;
7. interpretation, figure generation and report production.

The complete Python and shell code is provided in the accompanying repository. This document reads its principal tables and figures from the existing interpretation outputs and does not retrain models or regenerate predictions during compilation. Before the final clean rebuild, the audited timing, workflow and dependency issues described in [section 5](#) will be corrected.

B Additional Results

B.1 Model complexity

Table 2: Current model complexity

Model	Predictors	Nonzero	PLS comp.	Tree depth	Forest trees
Historical Mean	0.0000	–	–	–	–
OLS-3	3.0000	3.0000	–	–	–
PLS	499.0000	–	1.0000	–	–
Elastic Net	499.0000	0.0000	–	–	–
Decision Tree	499.0000	–	–	3.0000	200.0000
Random Forest	499.0000	–	–	–	–

B.2 Fixed versus optimized specifications

Table 3: Validation performance of fixed and optimized specifications

Model family	Fixed MSE	Optimized MSE	MSE improvement	ΔR^2_{OOS}
Random Forest	0.0080	0.0080	-1.3142×10^{-5}	0.0149
Elastic Net	0.0081	0.0080	0.0001	0.0523
Decision Tree	0.0085	0.0080	0.0004	0.0165
PLS	0.0082	0.0080	0.0001	–

B.3 Additional tree interpretation

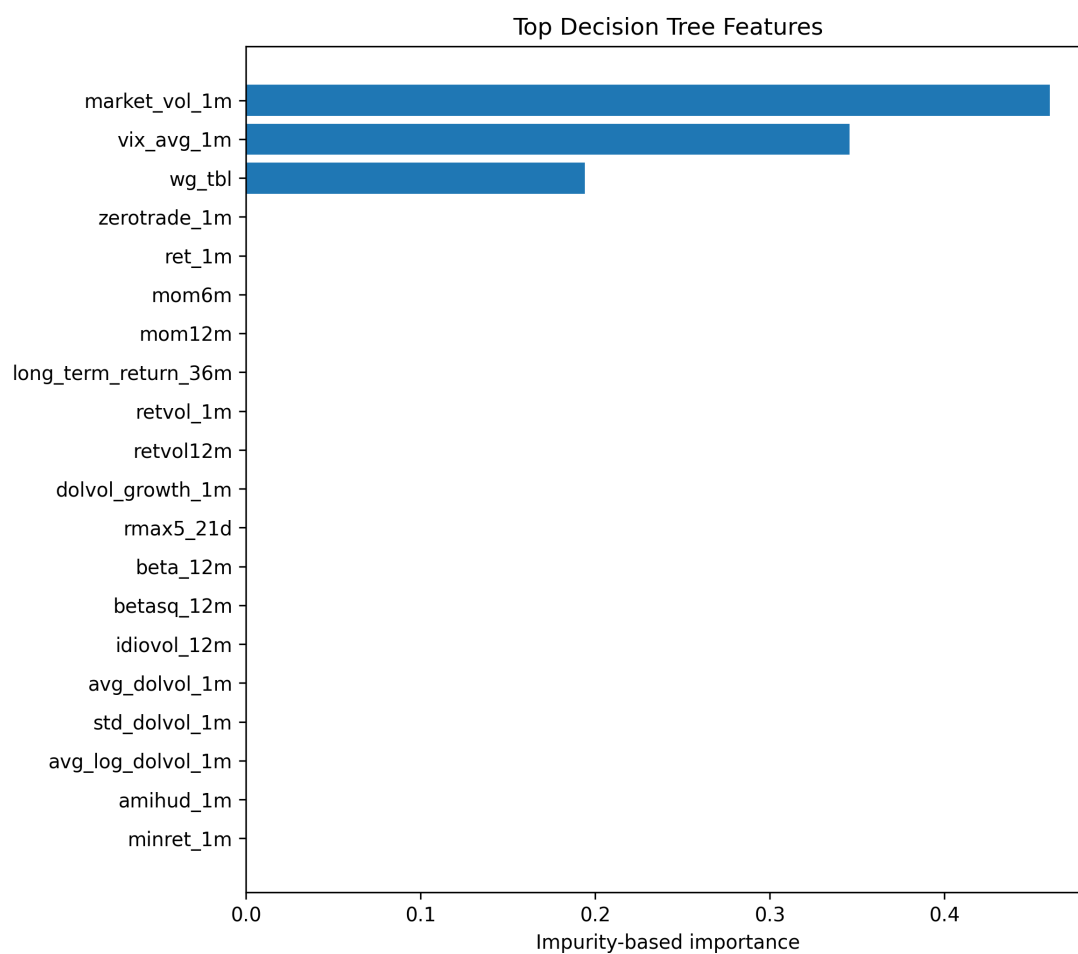


Figure 6: Top Decision Tree predictors based on impurity importance

The Decision Tree importance figure should be interpreted cautiously because the model has strongly negative out-of-sample performance. A variable can be important to a poorly performing fitted model without containing stable predictive information.